PROJECT DOCUMENTATION SUITE

CARBON HEIST

MITIGATION PLATFORM

NYC Local Law 97 Decarbonization Intelligence System

PROJECT TRACK Data Analysis & Engineering

DOMAIN Urban Sustainability, ML & Financial Risk Modeling

REPOSITORY github.com/ahmedadelamin/carbon-heist-mitigation

LIVE PLATFORM carbon-heist-mitigation.streamlit.app

VERSION v1.0 | July 2026

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1. Project Planning & Management

Project proposal, timeline, team structure, risk management, and success metrics.

11,638	81.65%	$268/MT	< 0.3s
BUILDINGS ANALYZED	MODEL ACCURACY	PENALTY RATE	RESPONSE TIME

1.1 Project Proposal & Executive Overview

About Local Law 97 (LL97)

NYC's LL97 imposes strict carbon limits on buildings over 25,000 sq.ft. starting 2024. Property owners exceeding thresholds face $268/MT CO_2e annually — fines that can reach millions.

The Carbon Heist Mitigation Platform is an end-to-end data engineering, machine learning, and financial decision-support system. It transforms raw municipal energy records into actionable engineering retrofits, CAPEX modeling, and accurate emissions forecasting.

Project Name	Carbon Heist Mitigation & NYC LL97 Decarbonization Intelligence Platform
Domain	Urban Sustainability, Data Science, Machine Learning & Financial Risk Modeling
Target Dataset	NYC Local Law 84 (LL84 Benchmarking) & Local Law 97 (LL97 Carbon Limits)
Core Formula	Penalty = Total Emissions × 268

Objectives

- **1. Automated Data Pipeline —** Build ETL pipeline ingesting 11,000+ properties with automated address standardization and anomaly detection.
- **2. Predictive Carbon AI —** Train Random Forest ML model (R^2 = 81.6%) to forecast GHG emissions and carbon liability ($/sq.ft.).
- **3. Database Standardization —** Design normalized schemas for MySQL/PostgreSQL and Microsoft SQL Server (T-SQL).
- **4. Executive Dashboard —** Real-time UI via Streamlit and Plotly to simulate decarbonization playbooks.

Project Scope

IN-SCOPE	OUT-OF-SCOPE
• Automated cleaning and anomaly detection on NYC Open Data. • ML regression modeling for emission forecasting. • Interactive dashboard for property owners and sustainability officers. • Multi-engine SQL relational database schemas.	• Real-time IoT hardware sensor integration. • Automated legal filing with NYC DOB.

1.2 Project Plan & Timeline

Phase	Key Deliverable	Start	End	Duration
Phase 1: Architecture	Requirements & Scope	May 12	May 18	7 Days
	Database Schema & ERD	May 19	May 25	7 Days
Phase 2: Data Pipeline	ETL Script & Cleaning	May 26	Jun 05	10 Days
	Audit Validation & Report	Jun 06	Jun 10	5 Days
Phase 3: AI Modeling	Feature Engineering	Jun 11	Jun 17	7 Days
	Random Forest Training	Jun 18	Jun 24	7 Days
Phase 4: Dashboard	Streamlit UI Development	Jun 25	Jul 01	7 Days
	Plotly Charts & Simulation	Jul 02	Jul 06	5 Days
Phase 5: Deployment	QA & Documentation	Jul 07	Jul 11	5 Days

1.3 Task Assignment & Roles

Team Role	Team Member	Key Responsibilities	Primary Deliverables
Lead Financial & Excel Strategist	**Ahmed Adel Amin**	Structure the master domain reference models (Excel Project/Co2 Project.xlsx) across all 16 sheets. Formulate statutory fine calculation ledgers ($268/MT) and blended payback formulas (7.58 Years). Design CAPEX self-funding reinvestment waterfalls and sensitivity matrices across the 5 engineering playbooks.	16-sheet master reference workbook (Co2 Project.xlsx); Itemized CAPEX/OPEX financial models & fine calculators.
Data Cleaning & Power BI Lead	**Ledia Sobhy**	Design and execute the automated ETL cleaning pipeline (data/Clean_Data_Pipeline.py). Handle missing feature imputation, address standardization, and statistical outlier filtering (Site EUI < 2000). Develop the standalone Power BI Executive BI Portal (NYC_Carbon_Heist_Mitigation_PowerBI_Dashboard.pbix) with custom DAX measures and 3-page boardroom visual analytics.	Cleaned master dataset (sample_nyc_energy.xlsx); Power BI Executive Portal (.pbix) & cleaning audit reports.

Team Role	Team Member	Key Responsibilities	Primary Deliverables
Visual BI & Tableau Specialist	**Hagar Hussein**	Design and package the progressive Tableau Executive BI Portal (NYC_Carbon_Heist_Mitigation_Tableau_Dashboard.twbx). Engineer multi-dimensional spatial treemaps and choropleth borough emission maps (Manhattan to Staten Island). Embed live mobile-scannable QR code portal linking directly to Tableau Public C-Suite executive views.	Tableau Packaged Workbook (.twbx); Spatial Choropleth Analytics & Live QR Portal.
Full-Stack App & ML Engineer	**Huda Amr**	Develop the interactive web presentation layer (application/app.py) using Streamlit and Plotly. Build 5 dedicated analytical tabs, custom KPI grids, and multi-slider retrofit simulation playgrounds. Train, tune, and evaluate the predictive Random Forest ML Regressor (R^2 = 81.65%) and serialize inference assets. Integrate Dual-Engine AI Co-Pilot & Chatbot (Gemini 2.5 Flash + local charting engine).	Interactive Streamlit Web Dashboard (app.py); Trained ML Regressor (1197_model.joblib) & AI Chatbot.
Relational Database Architect	**Abeer Adel**	Design normalized 3rd Normal Form (3NF) relational database architectures (PROPERTIES to LL97_PENALTIES). Write syntax-checked DDL scripts for dual database environments (carbon_heist_schema_mysql.sql & carbon_heist_schema_mssql.sql). Structure complete Entity-Relationship Diagrams (NYC_Energy_Chen_ERD.drawio).	Normalized SQL DDL schema scripts (MySQL & MSSQL); Complete relational ERD specifications.

1.4 Risk Assessment & Mitigation Plan

ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R-01	Dirty municipal data with typos and outliers.	High	High	Multi-step cleaning with regex normalization and outlier filtering (EUI < 2000).
R-02	Model overfitting on noisy anomalies.	Medium	High	Controlled tree depth (max_depth=20) and 80/20 train/test split.
R-03	SQL dialect incompatibility.	Medium	Medium	Separate DDL files for MySQL and Microsoft SQL Server.
R-04	Large file upload restrictions (~70MB).	High	Medium	Configured Git buffer (524MB) and lightweight serialization.

1.5 Key Performance Indicators (KPIs)

KPI Category	Target	Achieved Result	Status
ETL Data Retention	Retain 100% LL97 buildings (GFA ≥ 50,000) with 0% nulls.	11,638 validated records with complete integrity.	**PASS**
ML Accuracy	$R^2 \geq 0.75$ and MAE ≤ 250 MT CO2e.	R^2 = 81.65% and MAE = 212.99 MT CO2e.	**PASS**
Dashboard Response	UI recalculation under 1.5 seconds.	Sub-second (~0.3s) via Streamlit caching.	**PASS**

KPI Category	Target	Achieved Result	Status
Regulatory Precision	Exact: Penalty = Total Emissions × 268.	Verified across 13 engineering scenarios.	**PASS**

1.1.5 Strategic Decarbonization Financial Model

Portfolio-wide deployment of the 5 Strategic Decarbonization Playbooks across all 11,638 Local Law 97 audited properties — initial CAPEX, itemized recurring Annual OPEX ($15.70M/yr total), Gross Annual Savings ($656.63M/yr), Net Annual Benefit ($640.93M/yr), and payback horizons.

Playbook	Initial CAPEX	Annual OPEX	Gross Savings/yr	Net Benefit/yr	Payback	O&M Scope
01: Surgical Strike	$500,000	$45,000/yr	$20.59M	$20.55M	0.02 Yrs (~8 Days)	Level 2 audits, BMS scheduling & quarterly sensor calibrations.
02: Retro-commissioning	$802.17M	$3.25M/yr	$243.62M	$240.37M	3.29 Yrs	RCx ($1.50/ft²), continuous FDD software & semi-annual VAV tuning.
03: 1960s Smart Scale	$785.24M	$2.80M/yr	$88.03M	$85.23M	8.92 Yrs	Networked LED, VFD motors ($2.50/ft²) & preventative maintenance.
04: 1930s WET Systems	$1.50B (Net)	$4.50M/yr	$122.39M	$117.89M	12.25 Yrs	Wastewater heat recovery + 50% PPP grant ($749M), anti-fouling flushing.
05: Electrification Push	$1.89B	$5.10M/yr	$181.99M	$176.89M	10.38 Yrs	Electric heat pumps ($20/ft²), OEM service contracts & thermal scanning.
TOTAL PORTFOLIO	**$4.98B**	**$15.70M/yr**	**$656.63M**	**$640.93M**	**7.58 Yrs (Blended)**	**OPEX is 0.32% of CAPEX — 97.6% net recurring cash flow.**

2. Requirements Gathering & Stakeholder Analysis

Stakeholders, user stories, use cases, and functional/non-functional requirements.

2.1 Stakeholder Analysis

Stakeholder	Role & Interests	Pain Points	System Value
Asset Managers	Financial profitability and legal compliance.	Unbudgeted LL97 fines eroding NOI.	Real-time carbon liability ($/sq.ft.) and scenario modeling.
ESG Officers	Decarbonization strategies and Energy Star tracking.	Fragmented utility data; no predictive tools.	Automated pipeline and peer benchmarks.
MEP Engineers	HVAC, lighting, BMS, and retrofits.	Cannot identify highest ROI retrofit.	Engineering playbooks with ROI estimates.
Regulators	Compliance monitoring LL84/LL97.	Inconsistent reports and missing data.	Transparent audit reports and DB schemas.

2.2 User Stories & Use Cases

- **US-01 —** As an Asset Manager, I want to input building specs to instantly view projected LL97 penalty.
- **US-02 —** As an ESG Officer, I want to compare emissions intensity against peers to identify low-performers.
- **US-03 —** As a MEP Engineer, I want to simulate heating upgrades to justify CAPEX budget.
- **US-04 —** As a Data Analyst, I want automated cleaning for verified analysis.
- **US-05 —** As a C-Suite Executive, I want to converse with a dedicated AI Co-Pilot in Tab 5 to query portfolio breakdowns and dynamically generate custom Plotly charts on-the-fly without writing SQL or filtering rows.

ID	Name	Actor	Description
UC-01	Ingest & Clean LL84 Data	Data Engineer	Automated 8-step cleaning and validation.
UC-02	Predict Carbon Footprint	ML Engine	Predict GHG Emissions via Random Forest.
UC-03	Simulate CAPEX	Asset Manager	Adjust sliders to recalculate penalty exposure.
UC-04	Export ESG Report	ESG Officer	Generate KPI cards for C-suite executives.
UC-05	Conversational AI & Dynamic Charting	C-Suite Executive / Asset Owner	Query the 11,638-building portfolio via Tab 5 Dual-Engine AI Co-Pilot (Gemini 2.5 Flash + offline fallback) for instant natural-language insights and interactive Plotly charts.

2.3 Functional Requirements

- **FR-01 —** Ingest raw Excel (.xlsx) files per NYC LL84 standards.
- **FR-02 —** Replace "Not Available", filter < 50,000 sq.ft., correct city typos, remove meter gaps.
- **FR-03 —** Inference via Random Forest model for GHG emissions prediction.
- **FR-04 —** Calculate penalty: Penalty = Total Emissions × 268.
- **FR-05 —** Interactive KPI cards and playbook recommendations via Streamlit/Plotly across 5 specialized analytical tabs.
- **FR-06 —** SQL DDL structures for normalized property dimensions and compliance alerts.
- **FR-07 —** Dual-Engine Conversational AI & Dynamic Charting (Tab 5) integrating Google Gemini 2.5 Flash and a local offline quantitative fallback for on-the-fly interactive Plotly chart generation.

2.4 Non-Functional Requirements

- **NFR-01 Performance —** ML inference and UI recalculations under 1.0 second.
- **NFR-02 Scalability —** Batch processing of 50,000+ building records.
- **NFR-03 Reliability —** 100% deterministic financial calculations.
- **NFR-04 Usability —** Modern dark-mode design for executive scannability.
- **NFR-05 Cross-Platform —** Windows, macOS, Linux; MySQL and SQL Server.

3. System Analysis & Design

System architecture, database ER design, data flow diagrams, and process workflows.

3.1 Problem Statement

The Challenge

Raw disclosure data is often incomplete or noisy. Predicting future carbon penalties is extremely difficult. Without a predictive platform, managers risk excessive fines or overspending on retrofits.

Six-Pillar Dual-Engine Architecture (including AI Executive Chatbot)

Layer	Component	Key Files	Technology
1. Data Ingestion	ETL Pipeline	Clean_Data_Pipeline.py	Python, Pandas
2. Relational DB	Normalized 3NF Schema	carbon_heist_schema_*.sql	MySQL, SQL Server
3. Machine Learning	Prediction Engine	train_ll97_model.py, .joblib	Scikit-Learn
4. Presentation	Executive Dashboard	app.py	Streamlit, Plotly
5. Domain Knowledge	Financial Playbooks	Co2 Project.xlsx	Excel Modeling
6. AI Co-Pilot	Dual-Engine Chatbot	app.py (Gemini + Local)	Google Gemini 2.5 Flash, Plotly

3.2 Database Design & ER Diagram

The schema follows Third Normal Form (3NF) to eliminate redundancy:

Entity	Primary Key	Key Attributes	Relationships
CITIES	city_id	city_name (UNIQUE)	Referenced by BOROUGHS, PROPERTIES
BOROUGHS	borough_id	borough_name (UNIQUE)	FK: city_id → CITIES
PROPERTY_TYPES	property_type_id	property_type_name	Referenced by PROPERTIES
PROPERTIES	property_id	property_name, year_built, gfa	FK: borough_id, type_id, city_id
ENERGY_METRICS	energy_metric_id	energy_star_score, site_eui	FK: property_id
EMISSION_METRICS	emission_metric_id	total_ghg_emissions	FK: property_id
LL97_PENALTIES	penalty_id	base_penalty, penalty_per_ft2	FK: property_id
PROPERTY_FUEL_USAGE	fuel_usage_id	fuel_type, consumption_kbtu	FK: property_id

Entity	Primary Key	Key Attributes	Relationships
PROPERTY_ALERTS	alert_id	alert_type, description	FK: property_id

3.3 Data Flow & System Behavior

External Entity	Direction	Data Flow	Component
NYC Open Data Portal	INPUT	Raw Energy Benchmarking Records	Data Ingestion
Asset Manager	INPUT	Building Parameters & Sliders	Streamlit UI
System	OUTPUT	Projected Emissions & LL97 Penalties	ML + Financial Engine
System	OUTPUT	Decarbonization Playbook & ROI	Presentation UI

ETL Cleaning Workflow

Step	Activity	Description	Data Impact
1	Load Raw Data	Read LL84 Excel Spreadsheet	Full dataset loaded
2	Replace Nulls	"Not Available" → NaN	No row reduction
3	Filter Status	Remove test properties	Minor reduction
4	Filter GFA	Retain ≥ 50,000 sq.ft.	Significant filtering
5	Normalize City	Fix typos; enforce NYC only	Non-NYC removed
6	Filter Alerts	Drop meter gap records	Moderate filtering
7	Filter Outliers	Drop $/ft² > $1,700	Minor reduction
8	Deduplicate	Remove duplicate IDs	Minor reduction
9	Export	Save 11,638 clean records	Final dataset

4. Implementation, Coding Standards & Version Control

Coding conventions, modular structure, security practices, and version control.

4.1 Coding Standards

Python (PEP 8)

- **Variables & Functions —** snake_case for all variables, functions, and modules.
- **Classes —** PascalCase for class definitions.
- **Constants —** UPPER_CASE_WITH_UNDERSCORES for design tokens and thresholds.
- **Docstrings —** All core functions include explicit input/output documentation.

SQL

- **Keywords —** UPPERCASE (CREATE TABLE, PRIMARY KEY, FOREIGN KEY).
- **Table Names —** Plural uppercase nouns (PROPERTIES, EMISSION_METRICS).
- **Column Names —** Lowercase snake_case with unit suffixes (site_eui_kbtu_ft2).

4.2 Modular Code & Reusability

Directory	Purpose	Key Files
application/	Presentation Layer	app.py, input.xlsx, results.csv
data/	Data Engineering & ETL	Clean_Data_Pipeline.py, sample_nyc_energy.xlsx
database/	Persistence & DDL	carbon_heist_schema_mysql.sql, _mssql.sql, ERD
models/	AI & Prediction	train_ll97_model.py, ll97_playground.py, .joblib files
Excel Project/	Financial Domain	Co2 Project.xlsx (16 Sheets)
docs/	Documentation	6 Markdown files + Word document

4.3 Security & Error Handling

- **Data Validation —** Inputs wrapped in type coercion (float(), pd.to_numeric(errors="coerce")).
- **Outlier Guardrails —** Domain guardrails (EUI < 2000, GFA > 0) prevent bad predictions.
- **SQL Injection —** Parameterized schema definitions, no string concatenation.

4.4 Version Control Strategy

Platform	GitHub
Repository	github.com/ahmedadelamin/carbon-heist-mitigation
Primary Branch	main (production-ready)
Commit Convention	Meaningful atomic messages per feature update
Strategy	Single main branch with descriptive commits

5. Testing & Quality Assurance

Comprehensive test plan, automated validation, and bug resolution history.

5.1 Test Plan & Scenarios

ID	Category	Scenario	Expected	Actual	Status
TC-01	Data Eng.	Ingest 11,000+ raw records	"Not Available"→NaN	All records parsed	**PASS**
TC-02	Data Eng.	GFA ≥ 50,000 filter	LL97-subject only	11,638 retained	**PASS**
TC-03	Data Eng.	Borough normalization	0 invalid addresses	0 invalid found	**PASS**
TC-04	ML	Random Forest (80/20)	R^2≥75%, MAE≤250	R^2=81.65%, MAE=213	**PASS**
TC-05	ML	Filter parity	Both: EUI < 2000	Both: 81.6%	**PASS**
TC-06	Financial	LL97 fine calc	Emissions × 268	Exact match	**PASS**
TC-07	UI	Slider speed	Under 1.5s	Under 0.3s	**PASS**
TC-08	AI Co-Pilot / Tab 5	Query in English/Arabic ("Show top 5 emitting property types")	Parses intent, calculates totals, renders chart	Generated chart & text response	**PASS**
TC-09	AI Co-Pilot / Tab 5	API rate-limit shielding (HTTP 429 / no key)	Catches exception, routes to local fallback	Zero downtime, instant local chart	**PASS**

5.2 Automated Testing

Automated Data Integrity Checks

1. Assert cleaned dataset > 10,000 records. 2. Assert minimum GFA ≥ 50,000 sq.ft. 3. Assert GHG Emissions column has zero NaN values.

5.3 Bug Reports & Resolutions

Bug ID	Date	Description	Root Cause	Resolution
BUG-01	2026-06-30	Playground: 63.8% vs 81.6%.	Outdated filter (EUI<1500 vs EUI<2000).	Synced filters; restored 81.6%.
BUG-02	2026-06-30	SyntaxWarning: "\d" escape.	Windows backslash as regex.	Changed to POSIX forward slashes.
BUG-03	2026-07-02	MSSQL: ON DELETE RESTRICT error.	T-SQL unsupported syntax.	Created MSSQL schema with NO ACTION.

5.4 Enterprise Visual BI Verification Matrix (Power BI & Tableau)

Test ID	Component	Verification Procedure	Expected Outcome	Status
PBI-01	Power BI Data Binding	Open the Power BI dashboard offline without external database connection.	All 11,638 records load instantly from the internal data model without prompt errors.	**PASS**
PBI-02	Power BI Slicer Sync	Select Manhattan and Multifamily Housing on the left-hand slicers panel.	All KPI cards and charts across all 3 pages dynamically filter via synchronized DAX actions.	**PASS**
PBI-03	Power BI Waterfall Calc	Verify Page 3 (Carbon Cost & Savings Analysis) waterfall bridge values.	Exact parity between baseline penalty, savings across 6 playbooks, and residual liability.	**PASS**
TAB-01	Tableau Extract Binding	Open the Tableau dashboard offline without database connection.	All 11,638 records load instantly from internal .csv extract without prompts.	**PASS**
TAB-02	Tableau Borough & Type Filters	Select Manhattan and Commercial Office on global dropdowns.	All charts across all 3 pages dynamically filter via synchronized actions.	**PASS**
TAB-03	Tableau Fine Calc Accuracy	Audit Top Penalty Buildings bar chart against raw formulas.	Co-Op City output consistent with Python ML pipeline.	**PASS**
TAB-04	Tableau C-Suite QR Bridge	Scan embedded QR code from 2 meters with iOS/Android device.	Instant camera focus and redirection to the live Tableau C-Suite Portal.	**PASS**

6. User Manual, Deployment & Execution Guide

Installation instructions, execution commands, and interactive dashboard user guide.

6.1 System Requirements

Operating System	Windows 10/11, macOS 11+, or Linux (Ubuntu 20.04+)
Python	Python 3.10+
RAM	4 GB minimum (8 GB recommended)
Disk Space	500 MB free
Libraries	pandas, numpy, scikit-learn, openpyxl, streamlit, plotly, joblib, fpdf2

6.2 Installation Steps

Step 1 — Clone Repository

git clone https://github.com/ahmedadelamin/carbon-heist-mitigation.git → cd carbon-heist-mitigation

Step 2 — Install Dependencies

pip install pandas numpy scikit-learn openpyxl streamlit plotly joblib fpdf2

6.3 Execution Guide

Option	Command	Description
A: Dashboard	cd application && streamlit run app.py	Launch Streamlit at localhost:8501
B: AI Playground	cd models && python ll97_playground.py	Test building archetypes in terminal
C: Data Cleaning	cd data && python Clean_Data_Pipeline.py	Regenerate clean dataset

6.4 End User Manual

- **Step 1 — Configure Profile:** Use sidebar controls to select Borough, Property Type, Year Built, and GFA.
- **Step 2 — Tab 1 & Tab 3 Analysis:** Review real-time KPI cards, predicted GHG emissions (R^2 = 81.65%), liability ($/sqft), and peer gaps.
- **Step 3 — Tab 2 & Tab 4 Simulation:** Model HVAC retro-commissioning, envelope insulation, and electrification upgrades across the 5 Decarbonization Playbooks.

- **Step 4 — Tab 5 AI Chatbot:** Ask natural-language strategic questions in English or Arabic to dynamically generate custom interactive Plotly charts and quantitative audits (Dual-Engine: Gemini 2.5 Flash + Local Fallback).
- **Step 5 — Enterprise Visual BI Portals:** To complement the Streamlit dashboard and 16-sheet Excel model, the suite integrates standalone Dual-Engine 3-Page Executive BI Portals for Power BI and Tableau.

Power BI Portal

NYC_Carbon_Heist_Mitigation_PowerBI_Dashboard.pbix — features 3 progressive pages (Factors Affecting GHG Emissions → Asset Granularity & Construction Era → Carbon Cost & Savings Analysis Waterfall) built with Dark Royal Navy styling (#0F172A), custom DAX measures, and synchronized slicers for Borough, Property Type, and Decade Built.

Tableau Portal

NYC_Carbon_Heist_Mitigation_Tableau_Dashboard.twbx — features 3 progressive pages (NYC Energy Efficiency & Emissions Monitor → LL97 Compliance & Carbon Performance → Decarbonization Roadmap) packaged with internal data extracts (.csv) and an embedded mobile-scannable QR code framed in crisp white (#FFFFFF) for instant boardroom presentation bridging to the live online Tableau portal.

6.5 Database Initialization

MySQL / PostgreSQL	database/carbon_heist_schema_mysql.sql
Microsoft SQL Server	database/carbon_heist_schema_mssql.sql

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